

QP CODE: 4220100912



Reg No :

Name :

M.C.A. DEGREE EXAMINATION, JANUARY 2022

First Semester

Core - MCACT102 - "DIGITAL LOGIC & COMPUTER ORGANIZATION"

2020 Admission Onwards

5D53A44D

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Subtract 0012 from 1002.
2. Illustrate the use of error correcting codes.
3. Compare ASCII code and the EBCDIC code.
4. Map the expression $F = A'B'C + AB'C + A'BC' + ABC' + ABC$ on a Kmap.
5. Explain four input multiplexer with a neat diagram?
6. What is SISO shift register?
7. List the two basic memory operations with proper examples.
8. What is the role of MAR and MDR?
9. Explain why Cache access is faster than Main memory access.
10. Explain DMA transfer.
11. What are parallel computer structures?
12. Compare between shared and distributed memory.

(10×3=30 marks)

Part B

*Answer **all** questions*

*Each question carries **9** marks*

13. a) Explain the rules for the following (i) binary Addition; (ii) Binary Subtraction.

OR





- b) Explain the working of 7 bit hamming code with an example.
14. a) Reduce the Boolean expression $[AB' (C + BD) + A'B'] C$ as much as possible.
- OR
- b) Explain the four types of shift registers.
15. a) With a diagram, explain the components of a computer.
- OR
- b) Discuss in detail about the hardwired control unit with block diagram.
16. a) Explain Cache memory and its advantages.
- OR
- b) Describe how interrupts are handled by the processor?
17. a) Explain the Flynn's classification architecture of parallel processing.
- OR
- b) Explain the interconnection structures used in multiprocessors .

(5×9=45 marks)

